Silent but deadly? Physics says seldom, memory says mostly

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Abstract

The belief that a silent passage of flatus smells worse than an audible one appears in many languages but has not been tested. We follow a passage from its chemistry through sound and dispersion to a bystander's memory, combining measured odorant data, a lip-reed model of the anal valve, analytic and computational models of dispersion, and models of what an observer notices.

Methanethiol carries 43–94% of the summed odour activity of the three sulfur odorants, never less than hydrogen sulfide. Composition, not temperature, sets whether a passage rises or sinks, and gas-phase oxidation removes at most 0.4% of any odorant within two minutes in an office, so the odorants travel unchanged with the air. In the lip-reed model, 85% of releases in the measured range of passage volumes never set the valve oscillating; they are physically silent rather than quiet.

We prove that while transport is linear in concentration, a quieter release of the same gas cannot produce a higher peak concentration. Near the source, a loud jet is diluted before it slows to the speed of the surrounding air, by a factor proportional to the ratio of exit to ambient velocity, which we call the Silencer number. At equal mass this pre-dilution lets a silent release reach a neighbour's nose at a median 1.45 times the concentration of an audible one, often below the just-noticeable ratio of 1.28.

Silent passages are also smaller, and as emitted a silent passage is the stronger one at the neighbour in 48% of pairs.

In computational fluid dynamics of real scenes, the warm air rising along the body carries a release to different noses depending on its momentum. On a sofa the silent release rises behind the releaser's back to the ceiling and reaches neither nose within 30 s, while the audible one reaches both. In an elevator the silent release is stronger at every nose when a person stands behind the releaser and weaker at every nose when a wall does. Across these scenes a silent passage as emitted is the stronger one at a bystander's nose in 0–70% of pairs (mean 21%).

A silent passage can be noticed only by smell, so every silent passage a bystander remembers was smelled, while audible passages are remembered whether or not they were. With literature models of untrained observers, a remembered silent passage seems the stronger one in 68–93% of comparisons, against 32–51% over all passages. The releaser, who notices every release, is not subject to this selection.

1 Introduction

In the film *3 Idiots*,[1] a student nicknamed "Silencer" produces silent but foul flatulence and, in one scene, declares the silent release the deadliest. The belief behind the joke is old. An 1811 slang dictionary defines a related term as an escape "more obvious to the nose than ears",[2] and English has called such releases "silent but deadly" since at least 1970. A documentary survey of 32 languages (Section S10) finds the distinction between silent and loud releases lexicalised in at least five language families. The claim that the silent release smells worse is stated explicitly in Chinese and English sayings, and it is built into the definitions of smell-specific "silent" words that arose independently at least four times within Indo-European.

We found no test of the claim. The odour of flatus is well characterised: hydrogen sulfide, methanethiol and dimethyl sulfide carry it,[3,4] and the volume of a passage does not

predict its sulfur content.[5] Its sound has been attributed to vibration of the anal skin, like the buzzing lips of a brass player,[6] but no peer-reviewed study has measured it. The only data relating sound to smell are informal and not peer reviewed: in 1 619 recorded passages, louder ones were very slightly smellier by a gas-sensor reading ($R^2 = 0.016$), and none was silent.[7] Their direction agrees with what we find, but they cannot test a claim about silent passages and human noses.

The question is narrow, but answering it needs three results that are not specific to flatus.

- *Where does a release next to a body go?* The warm convective boundary layer around a person rises along the body to the breathing zone[8,9] and carries pollutants released near the body with it.[10] Near-body sources therefore give far higher exposure per unit mass than sources elsewhere in the room,[11,12] and CFD can follow particles shed from skin and clothing in this flow.[13] These studies concern passive sources and the occupant's own exposure. We show that the momentum of a release decides whether the boundary layer takes it, and so which of several nearby noses it reaches.

- *Can a slower release give a higher peak?* We prove that under linear transport it cannot, in any room, posture or flow. A slower release can win only in the non-linear near field, by a factor bounded by a velocity ratio.

- *How can a sensory belief arise that the physics does not support?* When an event can be noticed through either of two senses, selecting the noticed events makes the two senses appear related (Berkson's bias[14,15]). We show that this selection alone can explain the belief in a population where it is false, and we derive predictions that can test this explanation.

Both halves of the claim are also ambiguous. A release can be silent because it never sets the anal valve oscillating, or because background noise masks it. "Deadlier" can mean a higher peak at a nose, a larger total dose, or a stronger memory. The comparison can be between the same gas released in two ways (equal mass), between passages as they are

actually emitted, or between the passages an observer happens to notice. These bases give different answers, and we keep them apart throughout (Section 2.1).

We follow one passage from its chemistry through sound and dispersion to memory (Figure 1). Chemistry fixes which odorant carries the smell, what sets the buoyancy of a passage, and whether the odorants survive the trip to a nose. A lip-reed model of the valve gives the share of releases that are physically silent. Two theorems bound what linear transport can do to the silent/audible ratio; an analytic model of starting jets and puffs, checked against canonical jets and body plumes, gives the ratio over the range of releases; and computational fluid dynamics (CFD) of two people on a sofa, four in an elevator and two standing face to face shows where the body's own airflow takes each release. A third theorem and literature models of observers give how often a remembered silent passage seems the stronger one.

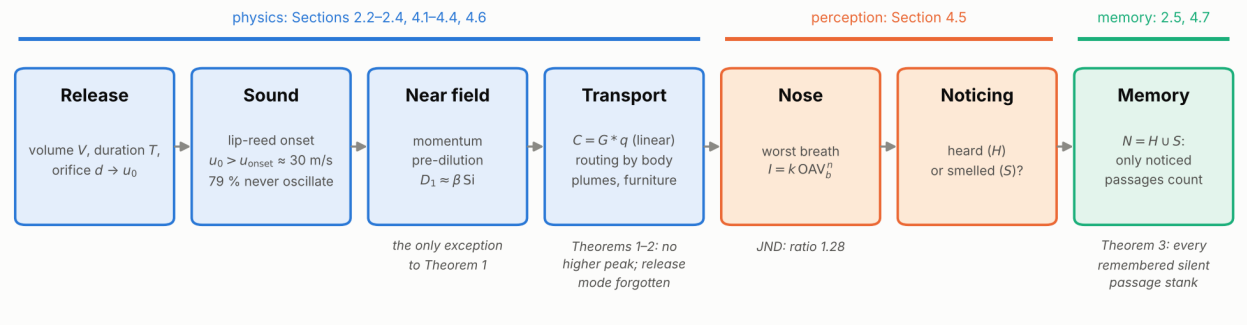


Figure 1: From release to memory: the quantity each stage passes on, and where Theorems 1–3 apply.

2 Definitions and theory

2.1 What "deadlier" means

A passage is a release of volume V over a duration T through an effective orifice of diameter d, with exit velocity $u_0 = 4V/(\pi d^2 T)$. We take its *deadliness* for a given person to be the

perceived intensity of the worst single inhalation,

$$I = k\,\mathrm{OAV}_{\mathrm{b}}^{n}, \qquad \mathrm{OAV}_{\mathrm{b}} = \max_{\text{breaths}} \frac{\int_{\text{insp}} \dot{V}_{\text{insp}}(t)\,\mathrm{OAV}(t)\,\mathrm{d}t}{\int_{\text{insp}} \dot{V}_{\text{insp}}(t)\,\mathrm{d}t}, \tag{1}$$

where $\mathrm{OAV} = \sum_i c_i/c_{\mathrm{th},i}$ is the summed odour activity value of hydrogen sulfide, methanethiol and dimethyl sulfide at the nose and n is the Stevens exponent. The breath is the natural unit: perceived intensity saturates within one natural sniff,[16] still integrates within it,[17] and the sniff is the unit of olfactory processing.[18] Cumulative exposure is reported as a secondary metric.

Every comparison below states its basis:

(A) *equal mass*: the same gas, released quietly or loudly;

(B) *as emitted*: silent and audible passages drawn from their own volume distributions;

(C) *as remembered*: only the passages an observer noticed.

Folk experience is basis (C). Physics answers (A) and (B).

2.2 Linear transport bounds the silent advantage

Odorants at sub-ppm levels are a passive scalar: the air carries them without being affected by them. Once the release has lost its own momentum and buoyancy, the concentration at a fixed nose is a linear functional of the release history $q(t)$ (volume of flatus per unit time):

$$C(t) = \int G(t-s)\,q(s)\,\mathrm{d}s = (G * q)(t), \qquad \int q\,\mathrm{d}s = V, \tag{2}$$

where the Green's function G encodes everything about the room, the bodies, the furniture and the ventilation. Time invariance of G requires the flow to be statistically stationary over the release, which holds for the quasi-steady convective boundary layers and room flows considered here.

Theorem 1. Let a silent release be a smoothed version of an audible one, $q_{\mathrm{s}} = q_{\mathrm{a}} * \kappa$ with $\kappa \geq 0$ and $\int \kappa = 1$. Then, for any G:

(i) the time-integrated dose $\int C \, \mathrm{d}t = V \int G$ is the same for both;

(ii) the peak concentration of the silent release does not exceed that of the audible one.

The proof (Appendix A) needs nothing about G beyond linearity, and it covers the breath-averaged OAV of Eq. (1) and releases stretched by any integer factor. In the linear regime, therefore, a quiet release can never be deadlier than a loud release of the same gas, whatever the posture, room or flow.

2.3 Release mode is forgotten

Theorem 2. In a closed, mixed room of fluid volume V_{room}, $G(t) \to 1/V_{\mathrm{room}}$ as $t \to \infty$. Both release modes therefore approach $C = V/V_{\mathrm{room}}$, and their ratio tends to one. More generally, any difference between release modes is carried by the early, near-field part of G, and it decays with travel time and distance.

2.4 Where the theorems fail: momentum and buoyancy

Equation (2) fails only near the source, where the release still carries its own momentum and buoyancy and so changes the flow that carries it. Two effects survive this.

Near-field pre-dilution. A momentum-dominated release entrains ambient air until its self-induced velocity has decayed to the ambient velocity u_h (the body's convective boundary layer or room air). For a starting jet, a puff or a steady jet alike, the dilution at that hand-off is

$$D_1 \approx \beta \, \frac{u_0}{u_h} \equiv \beta \, \mathrm{Si}, \qquad \beta = O(1) \tag{3}$$

(Appendix B). We call $\mathrm{Si} = u_0/u_h$ the *Silencer number*. An audible release (in the analytic model $\mathrm{Si} > 280$) arrives pre-diluted; a silent one ($\mathrm{Si} \approx 5$) does not. At equal mass, pre-dilution favours the silent release by at most the ratio of the two Silencer numbers.

Routing. Momentum also sets which flow the release joins, and different flows lead to different noses. This effect can favour either release. In the rooms of Section 4.4 it changes the ratio by more than pre-dilution does.

2.5 Selection biases what is remembered

Let H be the event that a passage is heard and S the event that it is smelled. An observer can only compare passages they noticed, $N = H \cup S$.

Theorem 3. Whatever the joint law of H and S,

$$P(S \mid \neg H, N) = 1, \qquad P(S \mid H, N) = P(S \mid H). \tag{4}$$

The proof is two lines (Appendix A). Among remembered passages, then, every silent one stank, while audible ones stank only at the base rate p_S, even when sound and smell are independent. The gap $1 - P(S \mid H)$ is largest when smelly passages are rare. This is Berkson's bias:[14,15] selecting on an event that either of two causes can produce (here, noticing, produced by hearing or by smelling) makes the two causes appear related among the selected cases. Nothing in it is specific to flatus: any event noticed through either of two senses is remembered with this bias.

An observer whose noticing does not depend on smell has no such bias. The emitter, who feels every release, has N certain for all passages and so compares silent and audible releases at their true frequencies.

3 Methods

Every parameter, with its source, range and whether it is measured, derived or assumed, is listed in Table S5. Code and data are archived (Section S11).

3.1 Source

Passage volumes span 17–375 mL, with a median of 92 mL.[5] In the Monte Carlo analysis, volume V, duration T (0.3–20 s) and effective orifice diameter d (1–8 mm) are drawn log-uniformly. Neither duration nor diameter has been measured for a passage; their upper bounds come from anorectal physiology (Section S2) and the lower bounds are assumed.

Sulfur gas concentrations are taken from 87 passages of 16 volunteers:[3] H_2S 1.06 µmol L^{-1}, methanethiol 0.21 µmol L^{-1} and dimethyl sulfide 0.08 µmol L^{-1}. They vary between passages by a log-normal factor with $\sigma_{\ln} = 1.1$, derived from the reported standard errors, drawn independently of volume because sulfur concentration does not correlate with passage volume.[5] Buoyancy is computed from measured compositions and is signed; it is drawn independently of volume because, across the 16 subjects of ref. 5, it does not correlate with mean passage volume (Spearman −0.15).

3.2 Odorants and sound

Details are in Sections S1 and S2. Each sulfur gas contributes an odour activity value, its mole fraction over its detection threshold; the passage's OAV is their sum (no mixture suppression). We compare two measured compositions[3,4] under three threshold sets, from a trained panel[19] to population means.[20] Buoyancy is split into a thermal and a compositional part, and gas-phase losses use IUPAC rate constants[21] with oxidant levels measured in an office.[22]

The anal canal is modelled as a lip reed, a soft-tissue valve that self-oscillates in the flow through it, as the lips do on a brass instrument.[6] It oscillates when the driving pressure $\frac{1}{2}\rho(u_0/C_\mathrm{d})^2$ exceeds a threshold of 0.6–2.5 kPa, bracketed by brass playing[23] and by oesophageal speech, the closest measured soft-tissue sphincter source.[24] This gives an onset exit velocity of about 42 m s^{-1} (21–71). An oscillating valve radiates as a harmonic monopole with a fundamental of 160 Hz (60–300 Hz). A release counts as *audible* when a harmonic exceeds both the hearing threshold[25] and the masked threshold of the background, from a

35 dB(A) living room to a 77 dB(A) subway car.

3.3 Analytic dispersion model

Dispersion is modelled in two phases. In the near field the release behaves as a starting jet and then a puff, or as a buoyant thermal, until its self-induced velocity has fallen to the ambient velocity u_h (Appendix B). The cloud is then advected to the nose and spreads as a Gaussian puff, $\sigma^2 = \sigma_0^2 + 2Kt$, with indoor eddy diffusivity $K = 10^{-4}$–$1.3 \times 10^{-2}\,\mathrm{m^2\,s^{-1}}$.[26] Two receptors are modelled: the releaser's own nose, reached by the convective boundary layer, the layer of warm air rising along the body,[8] at a speed taken from the CFD trajectories below; and a neighbour 0.8–1.4 m away, reached by room air. Concentrations are averaged over one inspiration.

We draw 20 000 pairs of silent and audible releases, each classified by the acoustic onset and evaluated with environmental parameters drawn from their ranges. Three simplifications favour the silent release: turbulent-jet constants are used even for laminar silent releases, the nose sits on the cloud centreline, and the centreline concentration is used.

3.4 Computational fluid dynamics

We use OpenFOAM 13 with transient Reynolds-averaged Navier–Stokes equations (RANS), the k–ω SST turbulence model, the Boussinesq approximation for buoyancy and the energy equation in internal-energy form.

Validation (Section S3). A free round jet at Re = 6700 reproduces the canonical decay constants (velocity 5.0, scalar $K_c = 4.3$–4.6) with a grid uncertainty of at most 3.7%. The convective boundary layer of a seated manikin is 40–56% faster than thermal-manikin data[8] and follows their trend with room temperature. Both biases favour the silent release: excess jet spreading over-dilutes audible releases by about 15%, and the fast boundary layer shortens travel times to the releaser's own nose by about 30%, which the analytic model corrects.

Scenes (Section S4). Manikins built from geometric primitives, with skin at 33 °C, stand or sit in rooms at 23 °C:

- a sofa: two seated people 0.6 m apart in a 3.2 × 3.2 × 2.6 m living room, with the releaser upright, reclined by 4.5 ° or leaning forward by 10 °, over 30 s (upright also 120 s);
- an elevator: four standing people in a closed 630 kg car, fan off, over 60 s, with the releaser in the front or the back row;
- a standing pair: two people face to face, noses 1.0 m apart, in the same living room, in still air and with office ceiling ventilation at 2.9 air changes per hour (EN 16798-1 Category II[27]), over 60 s.

Each scene is first run without a release until the body plumes have developed. From that flow field, the same 92 mL is released behind the releaser's buttocks, as an inlet that matches the release's volume flow and momentum: silent, 2 s through 8 mm at $u_0 = 0.9\,\mathrm{m\,s^{-1}}$; audible, 0.5 s through 2.4 mm at $41\,\mathrm{m\,s^{-1}}$. The odorant is a passive scalar, sampled 2.5 cm in front of each nose, over 5 cm discs at nose height and over sinusoidal breathing cycles.

3.5 Perception and observers

The nose samples only during inhalation: sinusoidal flow at 12 breaths per minute, inspiratory fraction 0.42.[28] Perceived intensity follows Stevens' power law, $I \propto \mathrm{OAV}^n$, with $n = 0.28$ measured for flatus H_2S.[29] Two passages are distinguishable when their concentrations differ by more than the olfactory Weber fraction of 0.28,[30] i.e. a just-noticeable ratio of 1.28.

The OAVs of the dispersion model use trained-panel thresholds.[19] For untrained observers:

- population thresholds are 10–23 times higher;[20,31]

- detection is a per-passage probability $\Phi(\log_2 \mathrm{OAV}/\sigma)$ with $\sigma = 2$ binary steps,[20] since one person's threshold varies over time about as much as thresholds vary across people;[32]

- a bystander who is not expecting an odour needs 26 times the threshold for a 50% response, with $\sigma = 3.5$ binary steps.[20,33]

Distraction reduces noticing[34] but not the peak rated intensity,[35] so intensity is computed from the attentive OAV. Two variants follow: clothing removes 22% of the sulfide gas,[36] and a sound can make a distracted bystander attentive for that passage.

3.6 Uncertainty analysis

Uncertainty is propagated through the analytic model; one CFD run of 30–60 s costs 1–3 h on four cores, too many for a sampling design. First-order Sobol indices (the share of the variance of $\log_{10} R$ explained by one input alone) are estimated by binning the Monte Carlo sample, with bootstrap 90% intervals. Because the silent and audible releases of a pair are drawn by rejection, the inputs are dependent and the indices are descriptive. The regime map shows $\log_{10} R$ at equal mass against the audible release's Silencer number and the travel time.

Passages as emitted in the simulated scenes. Each CFD scene gives the doses of one release volume at equal mass. The OAV is linear in concentration, so a passage of volume V following the same flow path gives OAV $\times\, V/92\,\mathrm{mL}$. For each of the 20 000 Monte Carlo pairs we scale the silent and the audible CFD doses by the pair's two volumes, and report $P(R > 1)$ and the shares above and below the just-noticeable ratio per scene and nose. In the analytic model this scaling is exact ($R_{\text{as emitted}} = R_{\text{equal mass}}\, V_s/V_l$). Applied to the CFD it assumes that the flow path does not depend on the volume: close for silent releases, whose momentum is negligible against the body plume, and an approximation for audible ones.

4 Results

4.1 The odorants and the gas that carries them

Over two measured compositions and three threshold sets, methanethiol carries 43–94% of the summed OAV and never less than H_2S (Section S1). This differs from the conclusion drawn in ref. 3, where H_2S is the main sulfur gas and the one whose concentration the judges' malodour ratings followed. Weighted by odour thresholds, the same measured concentrations give methanethiol at least an equal share: it is one fifth as concentrated, but its threshold is 5–6 times lower. Composition does not enter the silent/audible comparison, because transport is linear and acts on all three gases alike. Stevens' law with these OAVs predicts the judges' ratings of zinc-acetate-treated flatus (0.31–0.59 predicted, 0.28 and 0.59 rated) without fitting.

Composition, not temperature, sets most of the buoyancy: H_2 and CH_4 make flatus light and CO_2 makes it heavy, and after cooling to room temperature 25–38% of passages are heavier than air. In an office, gas-phase oxidation removes at most 0.4% of any odorant within 120 s, so the odorants travel with the air unchanged; surface deposition is unconstrained by data and would lower the doses of the silent release, which stays close to the body, further (Table S1).

4.2 Most releases are physically silent

In the lip-reed model, 85% of the releases sampled never set the valve oscillating (Figure 2); the other 15% are audible in any quiet room. No release in the sampled range oscillates without being heard in a quiet room: the smallest oscillating release, about 33 $\mathrm{mL\,s^{-1}}$ through a 1 mm orifice, gives a 47 dB fundamental at sofa distance (0.6 m), against an 18 dB hearing threshold at 160 Hz. A release that is not heard is therefore either physically silent or masked by background noise.

Masking needs loud backgrounds. In a restaurant or a subway car an oscillating release

must exceed 0.36–0.60 L s^{-1} at 0.6 m (1.8–3.0 L s^{-1} at 3 m) to be heard, so only near-maximal passages are: 2–4% of all releases at a 160 Hz fundamental, none at 100 Hz and 6–13% at 250–300 Hz. Audible releases are fast, with a median exit velocity of 90 m s^{-1} through a 1.5 mm orifice.

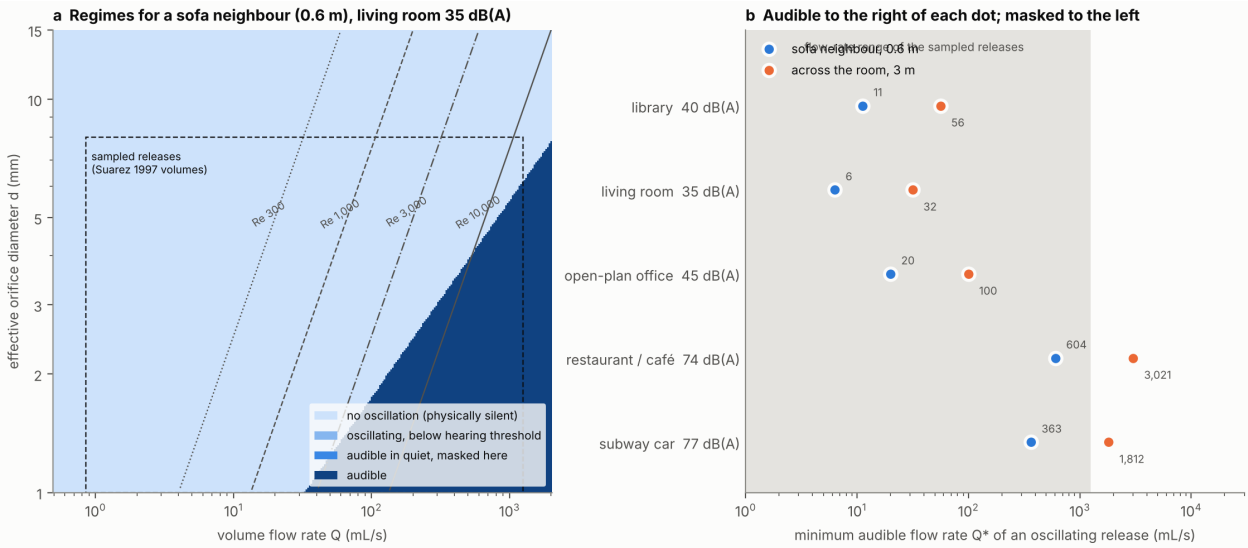


Figure 2: Acoustics. (a) Whether a release is heard, by volume flow rate and effective orifice diameter, for a neighbour at 0.6 m in a 35 dB(A) living room: the valve does not oscillate (physically silent); it oscillates but is masked here (audible in a quiet room); or it is audible. Dashed box: the sampled range of releases. Grey lines: jet Reynolds number. (b) Minimum audible flow rate of an oscillating release for five backgrounds, at 0.6 m and 3 m. The grey band is the flow-rate range of the sampled releases.

4.3 The ratio in the analytic model

Table 1 gives the breath-averaged ratio $R = C_{\text{silent}}/C_{\text{audible}}$ for 20 000 random pairs.

Table 1: Silent-to-audible concentration ratio in the analytic model (20 000 pairs).

receptor	basis	median R	5–95%	$P(R > 1)$
own nose	equal mass	1.94	0.94–16.4	0.93
own nose	as emitted	1.23	0.20–6.1	0.59
neighbour	equal mass	1.45	0.97–10.3	0.93
neighbour	as emitted	0.96	0.15–4.8	0.48
neighbour, denim	as emitted	0.62	0.11–3.3	0.31

Equal mass. At equal mass the audible release arrives pre-diluted by its own momentum (Eq. (3)), and the silent one is stronger at the neighbour in 93% of pairs. At the just-noticeable ratio of 1.28, the silent release is perceptibly stronger in 59% of pairs and indistinguishable in 39%.

As emitted. Audible passages are larger, and at the neighbour a silent passage is stronger in 48% of pairs: perceptibly stronger in 39%, perceptibly weaker in 42%. Trousers remove the jet momentum of both releases and lower $P(R > 1)$ to 0.31.

Which inputs matter. As emitted, the two passage volumes account for about 0.6 of the variance of $\log_{10} R$ at the neighbour; at equal mass, the eddy diffusivity K accounts for 0.37. Sphincter pressure, discharge coefficient, buoyancy and inhalation time each account for less than 0.5%: the valve parameters set whether a passage is heard, but hardly how strongly it smells.

Regime map (Figure 3a). The silent release is perceptibly stronger only at short travel times and high Silencer numbers of the audible release. At equal mass even a barely audible release is perceptibly weaker than a silent one, at the releaser's own nose (6.7 s; $R = 1.61$ at the acoustic onset) and at the neighbour (12.5 s; $R = 1.33$).

4.4 Real rooms: where the release goes

The analytic model carries every cloud straight to the nose. In the simulated rooms the warm air rising along the body takes the silent release with it, while the audible jet leaves it, and the two then reach different noses.

Sofa. The silent release rises in the gap between the releaser's back and the backrest, over the head and to the ceiling (Figure 4). Within 30 s its breathing-zone OAV stays at or below 10^{-10} at both noses. The audible jet hits the corner between seat and backrest, spreads sideways along the gap and reaches both breathing zones at OAV 0.2–0.6. With the releaser reclined by 4.5 ° or leaning forward by 10 °, the silent release again reaches neither breathing zone (at most 10^{-4}; Section S5). After about 70 s both clouds have returned from

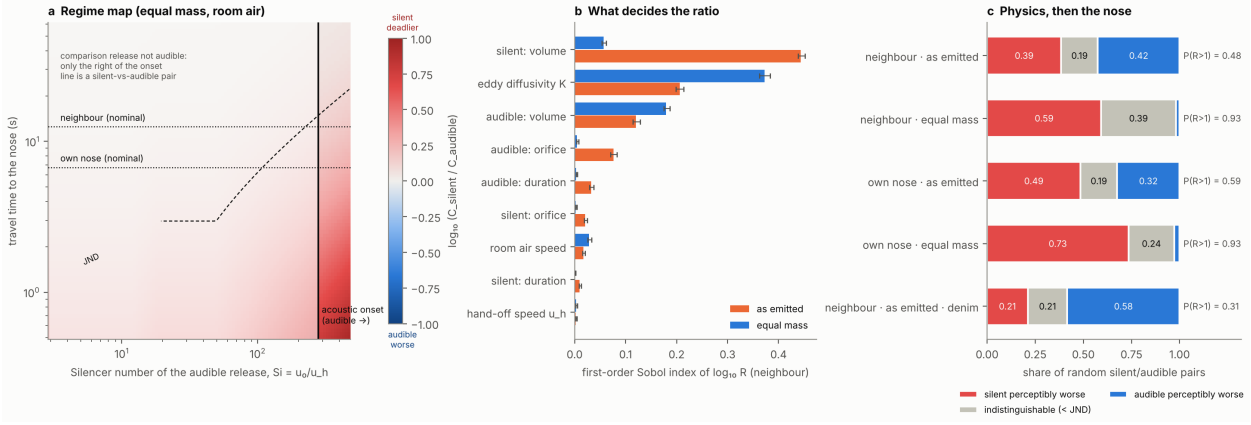


Figure 3: The ratio in the analytic model. (a) $\log_{10} R$ at equal mass and room-air mixing ($K = 10^{-3}\,\mathrm{m^2\,s^{-1}}$), over the Silencer number of the audible release and the travel time to the nose. The vertical line is the acoustic onset (Si = 279); to its left the comparison release is not audible (grey). The dashed contour is the just-noticeable ratio 1.28. Dotted lines mark the nominal travel times to the releaser's (6.7 s) and the neighbour's (12.5 s) nose. (b) First-order Sobol indices of $\log_{10} R$ at the neighbour, as emitted and at equal mass, with bootstrap 90% intervals. (c) Shares of random silent/audible pairs in which the silent release is perceptibly stronger, indistinguishable, or perceptibly weaker, with $P(R > 1)$.

the ceiling, and at the neighbour the two traces are indistinguishable (worst breath 0.39 for both), as Theorem 2 requires; at the releaser's own nose the mixed silent cloud is 1.5–2 times higher and touches the trained-panel threshold for about a second near 100 s.

Elevator. In a closed car with four standing occupants, the result depends on what stands behind the releaser. With a person behind (front row), the silent release rises in the 0.2 m gap into that person's breathing zone and is stronger at every nose ($R = 1.4$–4.3 at equal mass), reaching OAV 27 for the person behind. With the car wall behind (back row), it rises between back and wall, and the audible release is stronger at every nose ($R = 0.20$–0.55). With a front-row releaser the person behind reaches threshold first (8.8 s, against 17.6 s at the releaser's own nose), so the rule that whoever smells it first produced it holds for back-row releasers only. Onlookers who see only who reacts identify the releaser at 0.28–0.52, against 0.25 by chance (Section S6).

Standing pair. **[GAP: standing pair face to face, still air and office ventilation: results pending]**

Passages as emitted. Scaling each scene's equal-mass doses by the volumes of passages as emitted (Section 3.6) gives Figure 5. Silent passages are also smaller (median volume ratio 0.45), which lowers the silent advantage where the flow favours the silent release: for the person behind a front-row releaser, a silent passage is stronger in 35% of pairs. Over the nine bystander noses $P(R > 1)$ is 0–0.70, with an equal-weight mean of 0.21, against 0.48 in the analytic model.

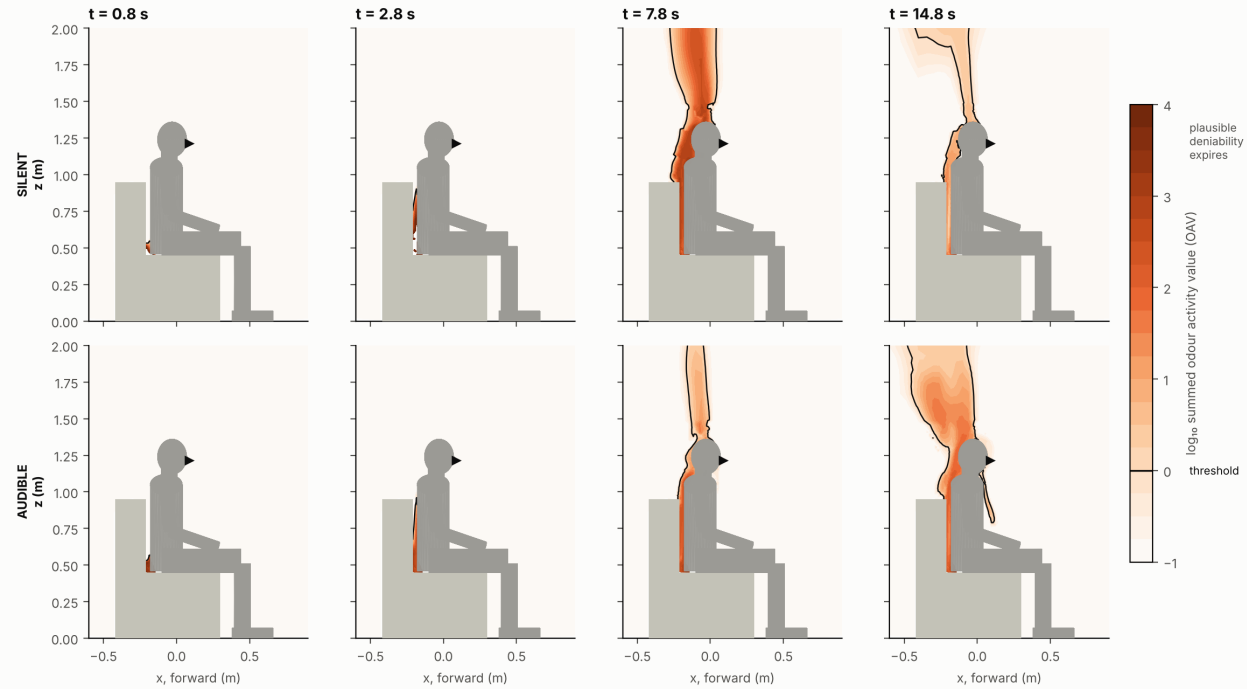


Figure 4: Sofa, releaser's mid-sagittal plane (upright): $\log_{10}$ OAV at 1, 3, 8 and 15 s after the release starts, silent (top) and audible (bottom). The black contour is the odour threshold.

4.5 Perception

Breath sampling. For a neighbour who is not warned, the worst breath still captures 81–99% of the peak. A listener warned by the sound needs about 25 s of breath-hold for an even chance of missing the cloud (Figure S6).

Silent versus audible. Through Stevens' law, the median physical difference at the neighbour becomes a 1% difference in perceived intensity, below the olfactory just-noticeable difference of about 7% in intensity.

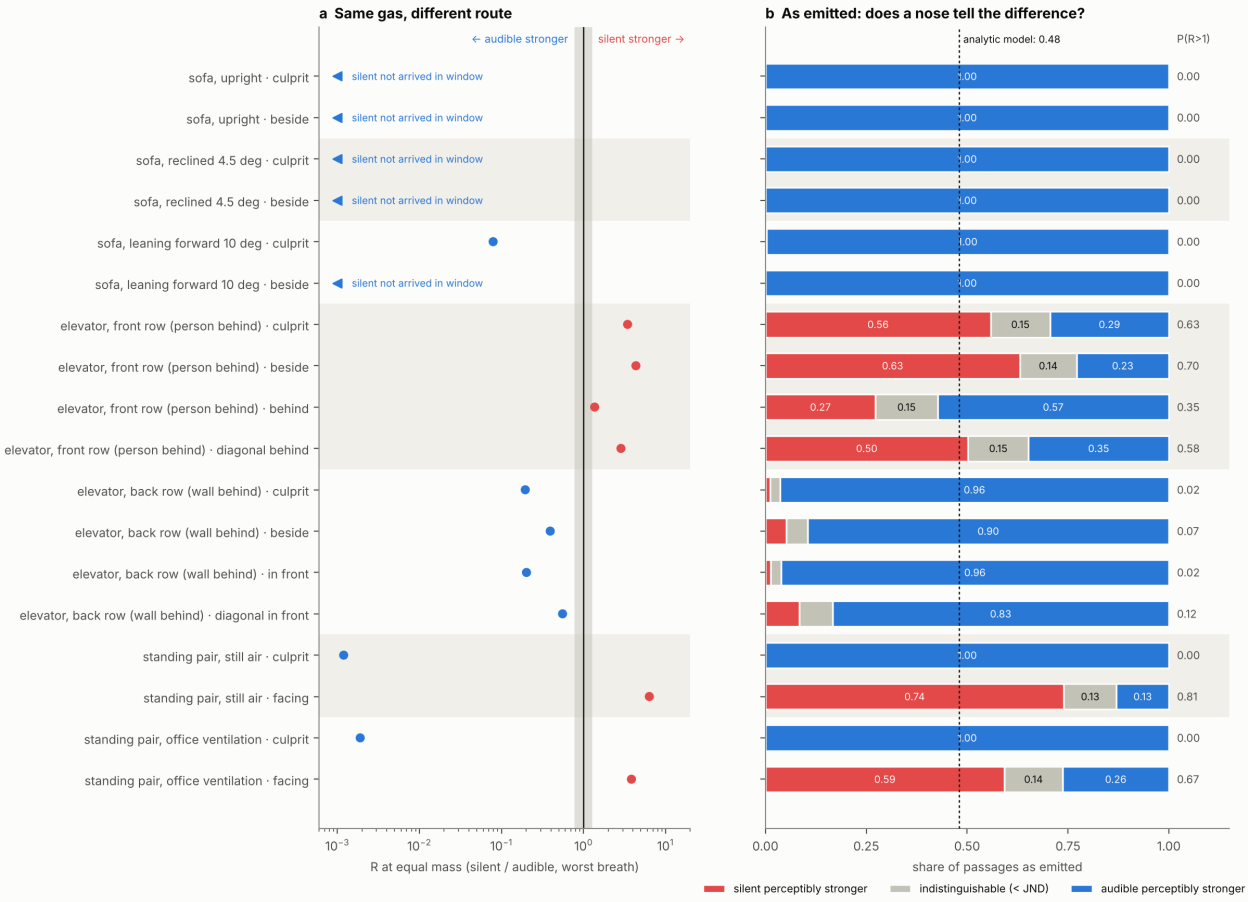


Figure 5: Silent versus audible in the simulated scenes. (a) Ratio of the expected worst-breath OAV (the most concentrated single inhalation) at each nose for the same 92 mL: sofa (30 s, three postures) and elevator (60 s, front-row and back-row releaser). Triangles: the silent release has not reached the nose within the window. The grey band is the just-noticeable ratio. (b) The same flow paths with passages as emitted (20 000 silent/audible volume pairs): shares of pairs in which the silent passage is perceptibly stronger, indistinguishable, or perceptibly weaker, and $P(R > 1)$. The dotted line is the analytic model.

Expectation. An expectation effect would have to exceed 1% in intensity to change which release seems stronger. Measured label effects for brief exposures are 0–3%, and their sign for an unwarned listener has not been established.[37–39] We do not include expectation in the model.

4.6 Memory

Applying Theorem 3 to the simulated passages gives Table 2 and Figure 6. Each base rate p_S assumes the cloud is centred on the neighbour's nose, so it is an upper bound.

Table 2: Share of passages smelled, p_S, and the remembered belief P(a silent passage seems stronger), for literature observer models; "truth" is the same probability over all passages.

observer	p_S	belief	truth
trained panel	0.93	0.53	0.49
attentive, population thresholds	0.59–0.68	0.68–0.73	0.50–0.51
distracted (everyday)	0.21	0.92	0.51
distracted, clothed releaser	0.19	0.93	0.51
distracted, sound cues attention	0.27	0.81	0.32

The fewer passages are smelled, the further the remembered belief departs from the truth. When a sound makes the bystander attentive, audible passages are smelled more often and the belief falls to 0.81; the truth for that observer is also lower (0.32), because they do smell audible passages more often. The releaser, who notices every release, shows no selection (Figure 6a). In loud backgrounds only the largest passages are heard, and this alone weakens the belief, so masking does not isolate the selection mechanism.

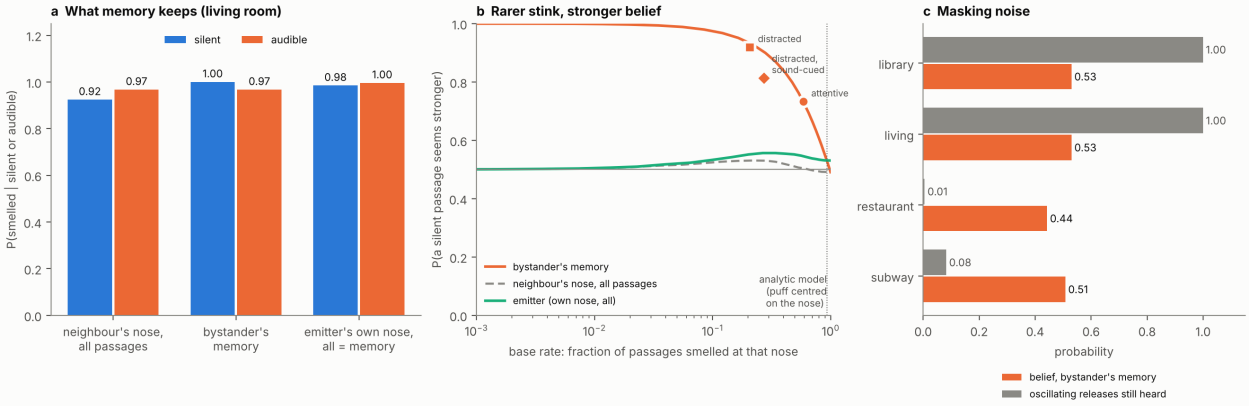


Figure 6: Memory, for 20 000 simulated passages. (a) Probability that a silent or audible passage was smelled: at the neighbour's nose over all passages, in a bystander's memory, and at the releaser's own nose. (b) Probability that a remembered silent passage seems stronger than a remembered audible one, against the share of passages smelled: bystander memory (orange), all passages (grey, dashed) and the releaser (green). Markers show the observer models of Table 2; the dotted line is the analytic model's nominal share. (c) Belief in four backgrounds, with the fraction of oscillating releases still heard.

5 Discussion

5.1 Why the belief exists

The physics we model does not make silent passages generally worse. At equal mass the silent advantage at a neighbour is a median factor of 1.45, often below the just-noticeable ratio. As emitted, a silent passage is the stronger one in 48% of pairs in the analytic model and in [**GAP: v4; preprint: 21% (0–70%)**] of bystander pairs in the simulated scenes, where on a sofa it misses both noses on its first pass.

The belief is nevertheless documented across languages. Theorem 3 can explain it without any physical difference: a passage enters memory if it is heard or smelled, so every remembered silent passage was smelled, whereas audible passages are remembered whether or not they smelled. With literature observer models this selection turns a truth of [**GAP: v4; preprint: 0.32–0.51**] into a remembered belief of [**GAP: v4; preprint: 0.68–0.93**] (Table 2). This shows that selection is sufficient for the belief, not that it is its cause; the predictions below can decide. Documented memory biases act in the same direction:

distinctive pairings are over-remembered,[40] and judgements built from biased samples are rarely corrected.[41] The phrase itself can also pass the belief on without any experience;[42] documentary evidence cannot separate the two routes (Figure S7).

5.2 Predictions

If selection is the cause, the belief should follow who notices what, and a survey can test this.

- *Releasers.* A releaser feels every release, so their noticing does not depend on smell and Theorem 3 does not apply to their own passages. Releasers should hold the belief less firmly than bystanders. Their own passages also disgust them less: the item "silent but smells strongly" is rated 2.51 for one's own flatus and 3.71 for a stranger's on a 1–5 scale.[43] This lowers the unpleasantness of their passages, not the contrast between silent and audible ones; only an expectation about their own odour[44] could carry the belief over to them.

- *People without a sense of smell.* As bystanders they can notice a passage only by hearing it, so they never remember a silent passage. If they hold the belief, it came from the phrase, not from experience.

- *Background noise.* Masking makes the quieter audible passages seem silent, so they enter memory only when smelled, but the passages still heard are the largest ones (Section 4.6). The two act in opposite directions, so the effect of noise on the belief should be ambiguous and is not a clean test.

5.3 Where the odour comes from

Two facts the transport model takes as given have chemical explanations in the literature, which we did not test (Section S9). The sulfur odorants are trace products held at a balance of production and fast mucosal consumption, whereas passage volume is set by the bulk

gases, which is why sulfur concentration and volume are uncorrelated.[5] Diet could shift this balance so that large passages carry less sulfur, in the direction of the belief; no study has measured sulfur gases and flatus volume together across diets.

5.4 When a silent release is worse

At equal mass, at the releaser's own nose, in still air and within a few seconds, a silent release is perceptibly stronger than a loud one (Figure 3a). In the elevator, a silent release is also stronger for the person standing behind a front-row releaser, who smells it first. In these two cases the saying holds.

5.5 Limitations

- *Mean flow only.* The CFD resolves the mean flow; instantaneous peaks of a meandering cloud can exceed the mean by an order of magnitude. This can lift the sub-threshold audible doses on the sofa (OAV 0.2–0.6) above threshold, but not the silent ones (10^{-10}). In the forward-leaning posture the mean silent cloud passes within 25 cm of the neighbour's face, where a filament could reach the nose.

- *Body plume.* The seated body plume is 40–56% too fast, which shortens the travel times of the silent release by about 30% in the seated case; the arrival times in the elevator (8.8 and 17.6 s) are likely short by a similar amount. Standing plumes were not validated separately.

- *Scenes and weights.* The as-emitted shares of Figure 5 weight the simulated scenes equally and assume the flow path does not depend on the passage volume. They describe these scenes, not how often each arises.

- *Base rate.* The share of passages smelled has never been measured; the remembered belief of [**GAP: v4; preprint: 0.68–0.93**] spans the observer models that bracket it.

- *Valve state.* The 85% silent share treats the onset pressure as a property of the tissue and does not distinguish flatus passing a relaxed sphincter from flatus passing a contracted one (Section S2).

6 Conclusion

- *Routing.* The body's convective boundary layer takes a slow release and lets a fast one pass, so the momentum of a release next to a body decides which nose it reaches. On a sofa the silent release rises behind the releaser to the ceiling and misses both noses on its first pass; in an elevator it is strongest for the person standing behind the releaser. [**GAP: v4: standing pair**]

- *Peak bound.* Under linear transport a slower release of the same gas never gives a higher peak, in any room or flow. Only the near-field pre-dilution of a fast jet lets a silent release win: by a median factor of 1.45 at equal mass in the analytic model, often below the just-noticeable ratio. As emitted, a silent passage is the stronger one at the neighbour in 48% of pairs in the analytic model and at a bystander in [**GAP: v4; preprint: 21% (0–70%)**] of pairs in the simulated scenes.

- *Selection.* A silent passage is noticed only if it is smelled, and this selection alone can explain the belief: in literature observer models a remembered silent passage seems the stronger one in [**GAP: v4; preprint: 68–93%**] of comparisons. The explanation predicts that releasers hold the belief less firmly than bystanders and that people without a sense of smell never remember a silent passage.

A Proofs of Theorems 1 and 3

Theorem 1. (i) follows from integrating Eq. (2) over t. For (ii), $C_{\mathrm{s}} = G * q_{\mathrm{a}} * \kappa = C_{\mathrm{a}} * \kappa$, and Young's inequality gives $\|C_{\mathrm{s}}\|_\infty \le \|C_{\mathrm{a}}\|_\infty \, \|\kappa\|_1 = \|C_{\mathrm{a}}\|_\infty$. □

A top-hat release lasting nT is a top-hat lasting T convolved with $\kappa = n^{-1}\sum_{j=0}^{n-1}\delta(t-jT)$, so stretching a release by an integer factor is covered. The breath-averaged OAV of Eq. (1) is itself a convolution of C with a non-negative kernel, so the same argument bounds it.

Theorem 3. On $\neg H$, noticing requires smelling, so $\neg H \cap N = \neg H \cap S$ and $P(S \mid \neg H, N) = 1$. On H, N is certain, so conditioning on N adds nothing. □

B Near field of the analytic model

The release behaves as a starting jet. Its tip follows $x = \Gamma(\dot{Q}_0 u_0)^{1/4} t^{1/2}$, and once the source stops it follows the puff law $x^4 = C_p^4 Q_0 u_0 t$, with $\Gamma = 2.8$ and $C_p = 2.6$.[45,46] The centreline dilution is $K_c d^*/x$ with $K_c = 5.11$. In parallel, a buoyant thermal grows with entrainment coefficient $\alpha = 0.25$,[47] and the larger of the two dilutions is kept. The near field ends when the self-induced velocity has fallen to the ambient velocity u_h (0.05–0.25 $\mathrm{m\,s^{-1}}$).

For a steady jet, a starting jet and a puff alike, the momentum carried by the cloud is conserved while it entrains, so the entrained mass grows as the inverse of the remaining velocity. At the hand-off velocity u_h the dilution is therefore proportional to u_0/u_h, which is Eq. (3); the prefactor β collects the jet or puff constants above and is of order one.

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Supporting Information Available

Overview diagram; CFD validation; the CFD scenes in detail (sofa postures, elevator, stand-

ing pair); breath sampling; the cross-cultural survey; source files of every result; and the full parameter table (PDF). Code and data: **[GAP: Zenodo DOI]**.

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TOC Graphic

